

MP491 2020/2021: Tutorial Sheet 5

Maps

I. Non-linear maps

Consider the following non-linear maps for $n = 0, 1, 2, \dots$. For each example find all fixed points and determine if they are stable or unstable.

1.

$$x_{n+1} = 2x_n - 2(x_n)^2$$

2.

$$x_{n+1} = \frac{1}{2}x_n - (x_n)^2 + \frac{1}{2}$$

3.

$$x_{n+1} = (x_n)^2 - x_n - 2$$

4.

$$x_{n+1} = x_n(x_n - 1)(x_n + 1)$$

II. The logistic map

Consider the one-parameter difference equation $x_{n+1} = ax_n(2 - x_n)$ obtained by the family of logistic maps

$$g(x) = ax(2 - x),$$

where a is a real positive parameter and $0 \leq x \leq 2$.

1. Investigate for what values of a the equation has **one** and **two** fixed points and investigate their stability.
2. Determine for what values of a the equation has period-2 points and what those points are. For what values of a are the resulting period-2 orbits stable and unstable?
3. Sketch, reasonably carefully, as much of the bifurcation diagram for the equation as you can. What happens as a passes through the critical value at which the period-2 orbits become unstable?